

## 1.7 Introduction to Proofs

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Def: A proof by contraposition of the conditional  $p \rightarrow q$  is a direct proof of the contrapositive statement  $\neg q \rightarrow \neg p$ .

Since  $p \rightarrow q \equiv \neg q \rightarrow \neg p$ , proving  $\neg q \rightarrow \neg p$  also proves  $p \rightarrow q$ !

Prove the statement...

"For real numbers  $x, y$ , if  $x + y \geq 2$  then  $x \geq 1$  or  $y \geq 1$ ."

$$\forall x, y (P(x, y) \rightarrow Q(x, y))$$

$$P(x, y) = x + y \geq 2$$

$$Q(x, y) = x \geq 1 \text{ or } y \geq 1$$

$$\neg P(x, y) = x + y < 2$$

$$\neg Q(x, y) = x < 1 \text{ and } y < 1 \text{ (De Morgan's)}$$

So we want to show  $\neg Q(x, y) \rightarrow \neg P(x, y)$ .

Let  $x$  and  $y$  be real numbers. Let

$x < 1$  and  $y < 1$ . Then

$x + y < 1 + 1 = 2$ . Therefore by proof by contraposition if  $x + y \geq 2$  either  $x \geq 1$  or  $y \geq 1$ .  $\square$

Def: A proof by contradiction of the statement  $p$ , is a direct proof that the conditional  $\neg p \rightarrow (r \wedge \neg r)$  is true for some proposition  $r$ .

"Prove that  $\sqrt{2}$  is irrational."

Suppose  $\sqrt{2}$  is rational. Then there are integers  $a, b$  such that  $\sqrt{2} = \frac{a}{b}$  where  $b \neq 0$  and  $a, b$

have no common factors.

By squaring this we find  $2 = \frac{a^2}{b^2}$  or  $2b^2 = a^2$ . By definition of even numbers,  $a^2$  is even. Since  $a^2$  is even,  $a$  is also even. Then there is an integer  $c$  such that  $a = 2 \cdot c$ . Using this in the previous equation gives  $2b^2 = (2c)^2 = 4c^2$  which simplifies to  $b^2 = 2c^2$ . By definition of even,  $b^2$  is even. So  $b$  is even. Then there is an integer  $d$  such that  $b = 2 \cdot d$ . We have found that

$$a = 2 \cdot c \text{ and } b = 2 \cdot d$$

so they have a common factor of 2. This contradicts our assumption that  $a$  &  $b$  have no common factors. Therefore it must not be the case that  $\sqrt{2}$  is rational, so  $\sqrt{2}$  is irrational. 

Proving "iff" Claims

"Prove that for integers  $m$  and  $n$ ,  $m^2 = n^2$  if and only if  $m = n$  or  $m = -n$ ."

Remember that the biconditional is logically equivalent to a conjunction of implications.

$$p \leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

So to prove an "iff" statement we must prove both directions.

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Let  $P(m, n): "m^2 = n^2"$  and

$Q(m, n): "(m = -n) \vee (m = n)".$

Then the claim may be expressed

as  $\forall mn (P(m, n) \Leftrightarrow Q(m, n))$

which is logically equivalent to

$\forall mn ((P(m, n) \Rightarrow Q(m, n)) \wedge (Q(m, n) \Rightarrow P(m, n)))$ .

So we must show both.

1)  $\forall mn (P(m, n) \Rightarrow Q(m, n))$

"If  $m^2 = n^2$  then  $m = -n$  or  $m = n$ ."

2)  $\forall mn (Q(m, n) \Rightarrow P(m, n))$

"If  $m = n$  or  $m = -n$ , then  $m^2 = n^2$ ."

Once this is done, we may conclude

"Therefore  $m^2 = n^2$  iff  $m = n$  or  $m = -n$ ."

Proof of 1)

Let  $m$  and  $n$  be integers. Suppose

$m^2 = n^2$ . (Want to show  $m = n$  or

$m = -n$ .) Then  $m^2 - n^2 = 0$ . This

is a difference of squares, thus

factors to be  $(m - n)(m + n) = 0$ .

For the product to be zero, one

of the factors must be zero so

$m - n = 0$  or  $m + n = 0$ . Then

either  $m = n$  or  $m = -n$ . Therefore

if  $m^2 = n^2$ ,  $m = n$  or  $m = -n$ .

Proof of 2)

Let  $m$  and  $n$  be integers.

Suppose  $m = n$  or  $m = -n$ . (Want

to show  $m^2 = n^2$ .) If  $m = n$ ,

then squaring both sides gives

$$\begin{matrix} 2 & 2 & + & r \\ & & & \end{matrix}$$

then squaring both sides gives  $m^2 = n^2$ . If  $m = -n$ , then

squaring both sides gives  $m^2 = (-n)^2 = n^2$ . Therefore if  $m = n$  or  $m = -n$ ,  $m^2 = n^2$ .

Therefore, for integers  $m$  and  $n$ ,

$m^2 = n^2$  only if  $m = n$  or  $m = -n$   $\square$

Prove that if  $x$  is rational and  $x \neq 0$  then  $\frac{1}{x}$  is rational.

$\Rightarrow$ ) Contradiction **What is P?**

Let  $x \neq 0$ . Suppose  $x$  is rational but  $\frac{1}{x}$  is irrational. Then since  $x$  is rational there are integers  $a$  and  $b$  with  $b \neq 0$  such that  $x = \frac{a}{b}$ . Since  $x \neq 0$ ,  $\frac{a}{b} \neq 0$ , so  $a \neq 0$ . Then  $\frac{1}{x} = \frac{1}{\frac{a}{b}} = \frac{b}{a}$ .

So  $\frac{1}{x}$  is a rational number.

This contradicts our assumption

that  $\frac{1}{x}$  is irrational. Then

it must not be the case that  $x$  is rational and  $\frac{1}{x}$  is irrational.

Therefore if  $x$  is a rational number, then  $\frac{1}{x}$  is a rational number  $\square$

$$(\sim(p \rightarrow q) \equiv p \wedge \sim q)$$

| P | q | $\sim q$ | $p \rightarrow q$ | $\sim(p \rightarrow q)$ | $p \wedge \sim q$ |
|---|---|----------|-------------------|-------------------------|-------------------|
| T | T | F        | T                 | F                       | F                 |
| T | F | T        | F                 | T                       | T                 |
| F | T | F        | T                 | F                       | F                 |
| F | F | T        | T                 | F                       | F                 |